Applied Surface Science 750 (2026) 167977

Contents lists available at ScienceDirect

Applied Surface Science

journal homepage: www.elsevier.com/locate/apsusc

Full Length Article

Ultra-high thermal boundary conductance in wafer-scale Si/diamond integration via surface-activated bonding

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ARTICLE INFO

Keywords:
Si wafer
Polycrystalline diamond
Surface activated bonding
Ultra-high thermal boundary conductance
Wafer-scale bonding
Low bow value

ABSTRACT

Diamond integration offers a compelling route to address thermal management challenges in advanced silicon (Si) electronics, yet remains limited by interfacial thermal resistance, process incompatibility, and poor scalability. Here, we demonstrate room-temperature surface-activated bonding (SAB) of polycrystalline diamond (PCD) to Si, achieving wafer-scale integration with ultralow bow. An ~ 18.5 nm amorphous interlayer forms at the as-bonded interface and transforms into a ~ 16 nm Si nanocrystalline layer upon annealing at 1000 °C, while maintaining a void-free and thermally stable interface. The thermal boundary conductance (TBC) reaches 133 MW/m^2 K, representing the highest Si/PCD TBC value reported to date in the literature. Scalable bonding up to 4-inch wafers with a bow of 30 ± 5 μm is demonstrated. These results reveal that controlled interfacial structural evolution enables simultaneous enhancement of thermal stability and interfacial heat transport, providing a viable pathway for next-generation high-power and high-density Si-based electronic systems.

1. Introduction

In the post-Moore's Law era, semiconductor devices are rapidly evolving toward higher power density and increased integration levels. As transistor scaling approaches the sub-2 nm node, the intensification of short-channel effects, along with the growing complexity of functional modules, has led to a substantial rise in power density and heat generation. Consequently, effective thermal management has become a critical bottleneck limiting the performance and reliability of silicon (Si)-based electronic systems [1–3]. It has been reported that for every 10 °C increase in operating temperature, the electrical performance of Si-based devices degrades by approximately 5%, while their service lifetime can be reduced by more than 50% [4]. In practical applications, high-performance chips—such as power devices and artificial intelligence processors—often operate at junction temperatures exceeding 100 °C, with power consumption approaching 100 W and local heat flux densities surpassing 1 kW/cm^2 [5,6]. These extreme thermal conditions push Si-based systems to their intrinsic physical limits. Meanwhile, conventional two-dimensional (2D) device architectures are no longer sufficient to meet the demands for high performance and low power consumption. Advanced integration technologies, such as 2.5D and 3D stacking, have emerged as promising solutions. However, the vertical stacking of multiple functional layers, combined with planar interconnect architectures, leads to highly confined heat flow pathways. This results in significantly elevated local heat flux densities compared to traditional planar systems, thereby further exacerbating thermal management challenges [7,8]. In addition, the relatively low thermal conductivity of Si (~142 W m^{-1} K^{-1}) limits its ability to efficiently dissipate Joule heat generated in active device regions, causing continuous temperature accumulation, performance degradation, and reduced reliability.

Diamond, owing to its exceptional electrical insulation, low dielectric constant, and ultrahigh thermal conductivity (~2200 W m^{-1} K^{-1}), has emerged as an ideal candidate for advanced thermal management in

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https://doi.org/10.1016/j.apsusc.2026.167977
Received 26 April 2026; Received in revised form 18 July 2026; Accepted 28 July 2026
Available online 29 July 2026
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microelectronic systems [9,10]. The integration of Si with diamond provides a promising route to overcome the thermal limitations of conventional Si-based devices. For instance, silicon-on-diamond (SOD) substrates have demonstrated significant potential for high-frequency and high-power applications by enabling efficient heat dissipation [11–18]. In advanced packaging, the incorporation of diamond as a heat spreader can effectively mitigate thermal accumulation, thereby improving device performance and reliability [19–22] Furthermore, heterogeneous integration with diamond also enables new device functionalities. For example, the formation of n^+/p^+-Si/p^+-diamond heterojunctions offers a viable approach to address the long-standing challenges of n-type doping in diamond [23,24]. In addition, such integration facilitates the planar co-integration of Si and diamond devices, which is particularly beneficial for optimizing the optical properties of diamond in quantum photonic applications [25].

Despite these advantages, the heterogeneous integration of Si and diamond inevitably introduces significant interfacial thermal resistance due to the low thermal boundary conductance (TBC), which limits the effective utilization of diamond's superior thermal conductivity. Currently, two primary approaches have been explored for Si/diamond integration: epitaxial growth and low-temperature direct bonding. In epitaxial growth, direct diamond deposition on Si substrates suffers from nucleation difficulties. The use of dielectric interlayers, such as SiN or AlN, can improve nucleation but inevitably increases interfacial thermal resistance [26–29]. Moreover, the initial diamond nucleation layer typically consists of small grains, resulting in reduced thermal conductivity [30]. Conversely, epitaxial growth of Si on diamond leads to high defect densities due to lattice mismatch, which degrades the crystal quality and device performance [31]. Although buffer layers can alleviate lattice mismatch, they introduce additional interfaces that further increase thermal resistance. In addition, epitaxial growth generally requires high temperatures, and the large mismatch in the coefficients of thermal expansion (CTE) between Si and diamond can induce severe wafer warpage or even cracking, severely limiting its compatibility with device fabrication.

As an alternative strategy, low-temperature direct bonding techniques —such as hydrophilic bonding (HB) [32,33], atomic diffusion bonding (ADB) [34–36], plasma-assisted bonding [37,38], and surface activated bonding (SAB) [39–42]—have been explored for Si/diamond integration. While HB method is effective to modulate surface wettability via solution-based treatment to enrich hydroxyl groups, forming well-bonded molecular interfaces upon annealing. However, the substantial disparity in hydroxyl adsorption density across different diamond crystal faces compromises interfacial bonding uniformity, and the stringent requirements on temperature and humidity confine this approach to single-crystal diamond, rendering it unsuitable for industrial-scale production [43]. To implement ADB, metallic interlayers (e.g., Ti, Au, Mo) are commonly deposited beforehand, inevitably giving rise to parasitic interfacial thermal resistance. Subsequent annealing activates sufficient cross-interface atomic interdiffusion and finally establishes a robust bonding interface between dissimilar substrates. Our previous studies have demonstrated the feasibility of SAB for heterogeneous integration of single-crystal diamond (SCD) and Si [44,45]. However, for practical and industrial applications, wafer-scale bonding is essential. Due to the high cost and size limitations of SCD, polycrystalline diamond (PCD) is considered a more viable alternative [46,47]. Nevertheless, PCD presents significant challenges for direct bonding. Its heterogeneous grain structure and varied crystallographic orientations make it difficult to achieve atomically smooth surfaces, which are required for high-quality bonding. In addition, the presence of grain boundaries and intergranular phases, such as amorphous carbon and impurities, leads to enhanced phonon scattering, thereby limiting interfacial heat transfer. To date, systematic studies on the direct bonding of Si and PCD remain scarce, and the fundamental mechanisms governing interfacial thermal transport are still not well understood. Moreover, a scalable integration approach that simultaneously achieves high TBC and low wafer bow has yet to be demonstrated, representing a critical barrier for practical device implementation.

In this work, we report the successful room-temperature SAB bonding of 1-inch PCD to 2-inch Si wafers. Furthermore, the scalability and robustness of this approach are validated by extending the bonding to 2-inch and 4-inch PCD wafers integrated with 4-inch Si wafers, achieving void-free interfaces and ultralow wafer bow. These results demonstrate that SAB provides a practical and scalable pathway for integrating PCD with Si, enabling high-TBC, low-distortion heterostructures for advanced thermal management in next-generation Si-based electronic systems.

2. Experimental

The PCD wafers were synthesized on poly-Si substrates with a thickness of 4 mm by microwave plasma chemical vapor deposition (MPCVD). Prior to diamond growth, the Si substrates were mechanically pretreated using fine sandpaper and diamond powder (50–100 nm in diameter) to introduce a high density of nucleation sites. Subsequently, the substrates were ultrasonically cleaned in acetone, ethanol, and deionized water for 5 min each to remove surface contaminants, followed by drying under high-purity N_2 before being loaded into the MPCVD chamber. The substrates were placed on a sample holder inside the chamber. After the base pressure was reduced below 1×10^{-5} Torr, hydrogen gas was introduced. A hydrogen plasma was generated using a microwave power of 2000 W and maintained for 15 min to further clean both the substrate surface and the chamber environment. Diamond growth was then initiated by introducing a CH_4/H_2 gas mixture with a methane concentration of 2–5%. The substrate temperature was controlled in the range of 700–1100 °C by adjusting the Z-axis position, thereby tuning the distance between the substrate and the plasma region. The gas flow rates were set at 10–25 sccm for CH_4 and 400–490 sccm for H_2. Different process parameters were employed to synthesize 1-, 2-, and 4-inch PCD wafers, leading to different deposition rates. The growth parameters for PCD wafers are summarized in Table 1.

After growth, the PCD-on-Si wafers were removed from the chamber and mechanically separated from the poly-Si substrates. However, PCD is formed by the competitive growth and coalescence of grains with different crystallographic orientations, containing abundant grain boundaries inside. Its surface exhibits uneven topography, which prevents intimate wafer contact during bonding and readily induces bonding voids. Fig. 1(a) shows the atomic force microscope (AFM) surface image of the 1-inch PCD wafer over a 20×20 μm^2 scanning area before chemical mechanical polishing (CMP) process, with an average surface roughness (Ra) roughness value of 3.46 nm. Therefore, the PCD surfaces were further processed by CMP before bonding. During the CMP process, H_2O_2 was used as an oxidizing agent to weaken the diamond C–C bonds through chemical oxidation, while SiO_2 and diamond abrasives were used to mechanically polish the surface layer. The process pressure was controlled at 0.3–0.5 MPa, while platen rotational speed and slurry flow rate were set at 20–30 rpm and 40–60 mL/min, respectively. Fine polishing was performed until the average surface roughness met the surface requirements for wafer bonding. Fig. 1(b) presents an 80×80 μm^2 AFM surface image of the 1-inch PCD wafer after polishing, where grain boundaries, intrinsic growth defects, and CMP-induced scratches are clearly resolved. The 1-inch PCD wafer exhibits an Ra of 0.95 nm, meeting the surface roughness criteria for wafer

Table 1
Growth parameters for PCD films with different wafer sizes.

Sample size	Microwave Power (KW)	Chamber Pressure (Torr)	Deposition Rate (μm/h)
1 in.-PCD	5.6	200	3.1
2 in.-PCD	7.8	130	2.3
4 in.-PCD	9.8	100	1.6

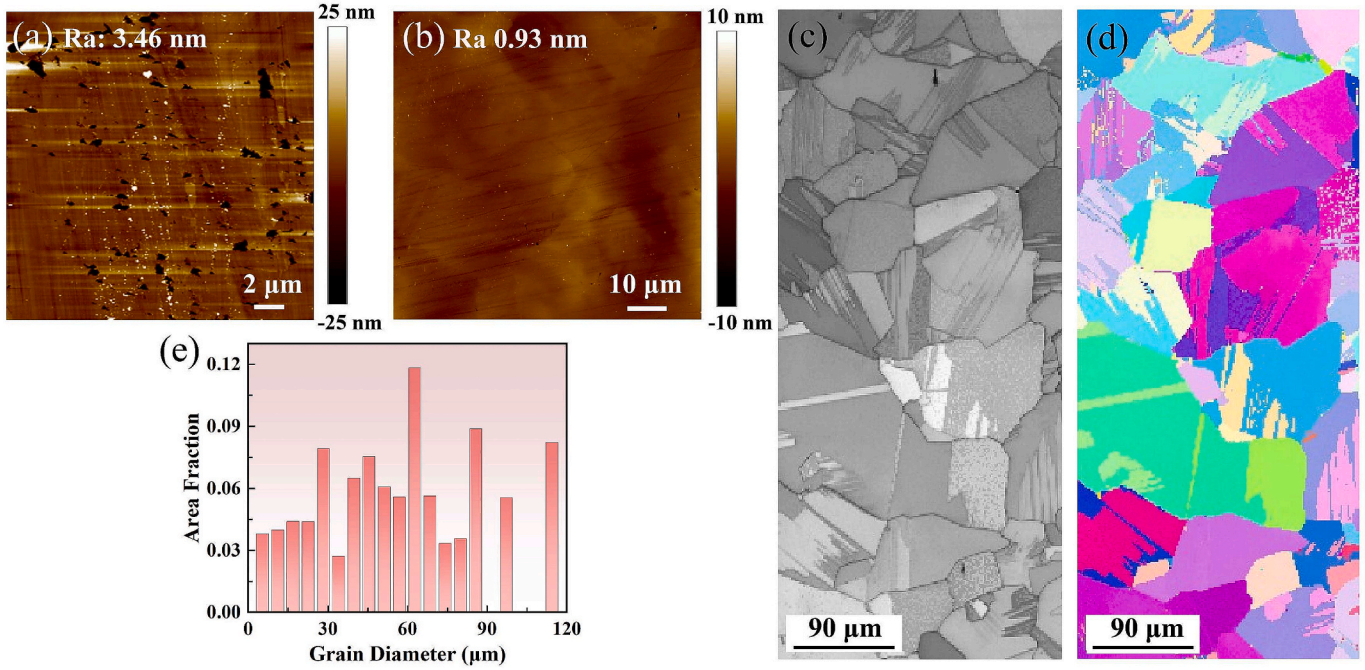


Fig. 1. The AFM surface images of PCD wafer surfaces before (a) and after(b) polishing; the EBSD image (c) and corresponding inverse pole figure (IPF) map (d) of the PCD wafer, showing the grain structure and crystallographic orientation; (e) the grain size distribution of the PCD wafer.

bonding. As shown in Fig. S1, a white light interferometer (WLI) was employed to measure surface roughness over an extensive area. The Ra values of the 1-, 2-, and 4-inch PCD wafers were measured to be 1.07, 0.63, and 0.58 nm over a wide 530 × 530 μm^2 scanning area, respectively. Subsequently, the polished PCD wafer surfaces were then cleaned using a sulfuric acid–hydrogen peroxide mixture to obtain a contamination-free surface suitable for bonding. The measured warpage values were approximately 9 μm, 16 μm, and 32 μm for the 1-, 2-, and 4-inch PCD wafers, respectively.The grain structure of the PCD was characterized using electron backscatter diffraction (EBSD). As shown in Fig. 1(c) and 1(d), the EBSD maps reveal well-defined grain boundaries with distinct crystallographic orientations. The corresponding grain size distribution (Fig. 1(e)) ranges from 10 to 115 μm, with a predominance of larger grains, which is beneficial for thermal transport due to reduced phonon scattering at grain boundaries [48]. Both the Si and polished PCD substrates were ultrasonically cleaned in acetone and ethanol for 300 s each, followed by drying under high-purity N_2. No additional chemical treatments were applied in order to preserve the intrinsic surface states. The cleaned wafers were then transferred into the SAB chamber. Surface activation was performed using an Ar fast atom beam (FAB) with a beam current of 1.8 mA and an acceleration voltage of approximately 2 kV for 180 s. Bonding was subsequently carried out under an applied pressure of 19.7 MPa, while maintaining an ultrahigh vacuum of ~ 1 × 10^{-7} Pa throughout the process.

The surface morphology and grain structure of the polished PCD wafers were characterized by AFM, WLI, and EBSD, respectively. The bonding quality and wafer bow of the bonded samples were subsequently evaluated using scanning acoustic microscopy (SAM) and photoluminescence (PL) spectroscopy. To gain further insight into the interfacial structure, high-resolution transmission electron microscopy (HR-TEM) was performed, enabling direct observation of the bonding interface as well as its evolution under high-temperature annealing. In addition, energy-dispersive X-ray spectroscopy (EDS) was employed to analyze the compositional and structural changes at the Si/PCD interface before and after annealing at 1000 °C. Finally, time-domain thermo-reflectance (TDTR) measurements were conducted to systematically quantify the TBC at the Si/PCD interface and its temperature dependence under different annealing conditions.

3. Results and discussion

Fig. 2(a) presents the heterogeneously integrated 1-inch PCD and 2-inch Si wafers fabricated by SAB, followed by annealing at 1000 °C. After annealing, a bonding area of approximately 80% was achieved, indicating that the bonded interface can withstand the stringent thermal conditions required for Si-based device processing. The scalability of the SAB approach is further demonstrated in Fig. 2(b) and 2(c), where bonding areas of ~ 99% were achieved for both 2-inch PCD/4-inch Si and 4-inch PCD/4-inch Si wafer-scale integrations. These results confirm the excellent scalability and manufacturability of the process, addressing a critical requirement for practical semiconductor applications.

SAM was employed to evaluate the bonding interface. As shown in Fig. 3(a–c), SAM images of 1-, 2-, and 4-inch PCD wafers bonded to 4-inch Si substrates reveal a significant reduction in the void ratio, decreasing from 64.19% (1-inch PCD) to ~ 1% (4-inch PCD). Supplementary Fig. S2 displays the SAM image of the 1-inch PCD/2-inch Si bonded sample detected from the Si side. The areas with color gradients in Fig. 3(a) correspond to incompletely bonded regions, which can be attributed to the non-uniform thickness of the diamond substrate. The PCD surface is consisted of numerous grains with distinct crystallographic orientations that coalesce to form grain boundaries, as illustrated in Fig. 1(d). These boundaries exhibit undulations with varying degrees distributed across the surface. Therefore, the formation of voids is primarily attributed to the intrinsic surface roughness and heterogeneous grain orientations of PCD, as illustrated in Fig. 3(d)–3(f). With improved surface morphology of the PCD wafers, the bonding quality was markedly enhanced, resulting in nearly void-free bonding for the 4-inch PCD/Si wafer, with a residual void ratio as low as ~ 1%. It was also observed that smoother surfaces were corresponding to larger effective bonding areas. Additionally, wafer surface cleanliness was identified to be a critical factor to govern bonding quality, and thus to achieve the void-free bonding of the 4-inch PCD/Si wafer. These factors may also be the reasons for achieving perfect bonding of 4-inch Si and 4-inch PCD wafers.

Fig. 4(a) shows the wafer bow mapping of a 4-inch PCD/4-inch Si bonded structure, exhibiting an ultralow bow value of 30 ± 5 μm, where the variation range is primarily determined by equipment-induced

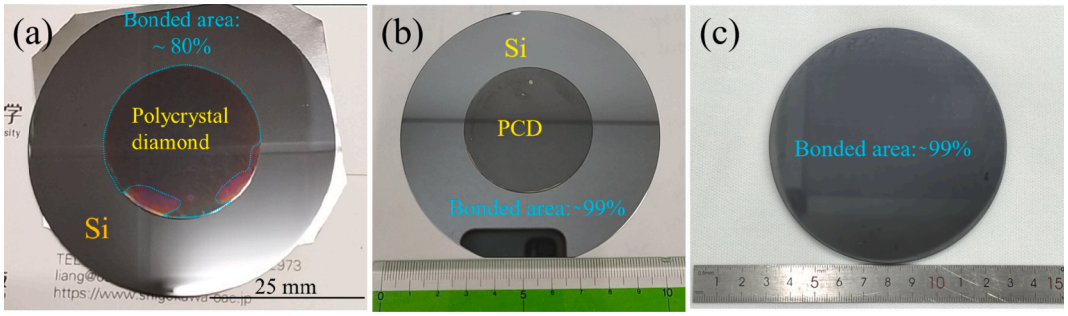


Fig. 2. Optical microscopy images of (a) a 2-inch Si/1-inch PCD bonded sample after annealing at 1000 °C; (b) a 4-inch Si/2-inch PCD bonded sample; and (c) a 4-inch Si/4-inch PCD bonded sample.

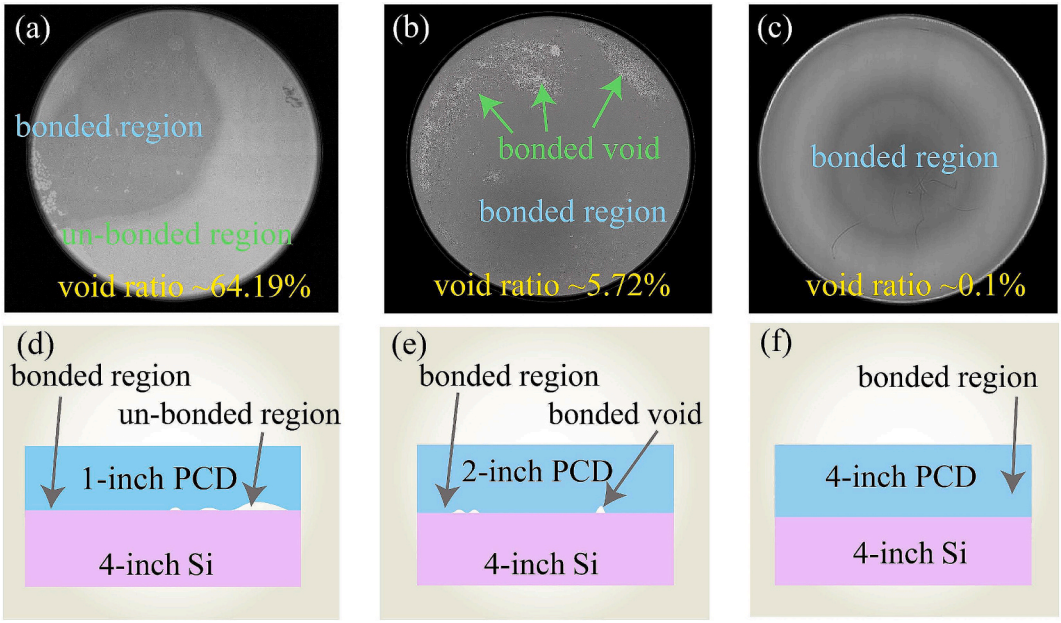


Fig. 3. SAM images of (a) a 2-inch Si/1-inch PCD bonded sample after annealing at 1000 °C; (b) a 4-inch Si/2-inch PCD bonded sample; and (c) a 4-inch Si/4-inch PCD bonded sample. Corresponding schematic illustrations of (d) the 2-inch Si/1-inch PCD, (e) the 4-inch Si/2-inch PCD, and (f) the 4-inch Si/4-inch PCD bonded configurations.

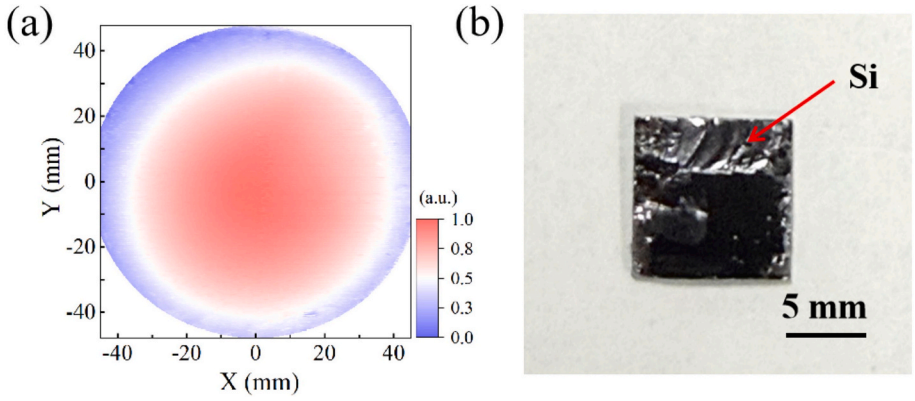


Fig. 4. (a) The bow mapping of 4-inch Si/4-inch PCD bonded wafer; (b) surface morphology photograph of the diamond side for as-bonded Si/PCD subjected to bonding strength push-shear test.

measurement errors. This significant reduction is attributed to the room-temperature SAB process, which effectively suppresses thermal stress arising from the mismatch in CTE and minimizes lattice-induced stress accumulation. Compared with conventional epitaxial growth of diamond on Si, the SAB approach achieves substantially lower wafer bow, representing a key advancement toward compatibility with standard semiconductor fabrication processes. Notably, the achieved bow value satisfies the tolerance requirements for Si-based device

manufacturing [49]. Fig. 4(b) presents the shear test result of the Si/PCD bonded sample on the diamond side. During the test process, the diamond side was clamped and fixed. A lateral thrust of 20 kgf was applied by a shear indenter perpendicularly against the edge of the silicon wafer at a position 200 μm above the bonding interface. It was observed that fracture occurred inside the silicon wafer rather than interfacial delamination at the bonding joint, demonstrating that the bonding strength exceeded the fracture strength of bulk silicon.

Fig. 5(a) and 5(b) present HR-TEM images of the Si/PCD bonding interface and its magnified view. An interfacial layer with a thickness of approximately 18.5 nm is clearly observed at the as-bonded interface, which is likely formed as a result of Ar FAB irradiation during the SAB process. Importantly, no cracks or interfacial voids are detected, confirming the formation of a uniform and mechanically robust heterointerface between Si and PCD. To further elucidate the interfacial structure, fast Fourier transform (FFT) analysis was performed, as shown in Fig. 5(c) and 5(d). The FFT patterns indicate that the Si region retains a well-defined crystalline structure, whereas the interfacial layer exhibits an amorphous nature, characterized by the absence of long-range atomic order. Such FAB-induced amorphous interlayers are commonly observed in surface-activated bonding processes and have been widely reported in various heterogeneous material systems, including GaN/diamond, Si/Si, Si/SiC, GaAs/SiC, and SiC/β-Ga_2O_3 [50–54]. This observation highlights the universal nature of ion-beam-induced amorphization in SAB and underscores its critical role in enabling strong bonding across dissimilar materials by accommodating lattice mismatch and promoting interfacial adhesion.

Fig. 6(a) and 6(b) present the EDS elemental mapping and the corresponding X-ray intensity profiles across the Si/PCD bonding interface. The spatial distributions of carbon (red), oxygen (green), silicon (blue), and argon (yellow) are clearly resolved, confirming the compositional heterogeneity of the bonded interface. In the as-bonded state, both oxygen and argon are detected within the amorphous interfacial layer. The presence of oxygen is attributed to residual contamination in the vacuum environment, while the incorporation of argon originates from the Ar FAB irradiation during the SAB process. In addition, carbon atoms are observed to diffuse into the interfacial layer, further indicating the formation of a chemically mixed region. These results reveal that the bonding interface is not atomically abrupt but instead consists of a compositionally complex amorphous layer formed during FAB activation. Such interfacial mixing is expected to influence both the structural stability and phonon transport across the interface, which will be further discussed in relation to thermal boundary conductance in the following section.

To evaluate the thermal stability of the bonding interface, HR-TEM images of the Si/PCD interface after annealing at 1000 °C in a nitrogen atmosphere are shown in Fig. 7(a) and 7(b), along with their magnified views. Despite the significant mismatch in the CTE between Si and PCD wafers, the interface remains free of cracks and voids after high-temperature annealing and subsequent cooling. This observation result indicates that the bonding strength is sufficient to withstand the thermally induced stress owing to the CTE mismatch. Following annealing, an interfacial layer with a thickness of approximately 16 nm is observed, slightly reduced compared to the as-bonded state. As shown in Fig. 7(c) and 7(d), lattice fringes corresponding to Si nanocrystals are clearly identified within this interfacial region, providing direct evidence of partial recrystallization from the initially amorphous layer. This microstructural evolution suggests a transition from a fully disordered structure to a partially ordered state, which contributes to the reduction in interfacial disorder. Such recrystallization not only decreases the thickness of the amorphous layer but also facilitates stress relaxation at the interface by accommodating lattice mismatch [45]. Moreover, the formation of nanocrystalline domains may provide more efficient pathways for phonon transport compared to the fully amorphous state. These results demonstrate the excellent thermal stability and mechanical robustness of the Si/PCD bonding interface, indicating its compatibility with high-temperature processing conditions required for Si-based device fabrication and heterogeneous integration.

Fig. 8(a) and 8(b) present the EDS elemental mapping and the corresponding X-ray intensity profiles of the Si/PCD bonding interface after annealing at 1000 °C. The elemental maps clearly reveal the spatial distributions of carbon (red), oxygen (green), silicon (blue), and argon (yellow) across the interfacial region. After annealing, pronounced interdiffusion behavior is observed at the interface. Specifically, oxygen and carbon atoms diffuse into the Si substrate adjacent to the interfacial layer, while silicon atoms counter-diffuse into the diamond side near the interface. Meanwhile, a residual fraction of Ar atoms remains at the interface, which can be attributed to the void-free nature of the Si/PCD bonded interface that prevents trapped Ar from surmounting the escape barrier even under thermal activation, thereby confining a small amount of Ar at the interface—a phenomenon commonly observed in SAB[55]. Nevertheless, Fig. 8(a) reveals that these trapped Ar atoms are sparsely and randomly distributed with a negligibly low overall concentration. Coupled with the intrinsically high thermal conductivity of diamond, the impact of residual Ar on the interfacial thermal conductance is slight. The density of residual Ar may be further reduced through subsequent process tuning. In addition, a noticeable enrichment of oxygen is detected within the interfacial layer, particularly in the region close to the Si substrate. It has been reported that a thin SiO_2 layer formed between PCD and Si creates a chemical gradient. Compared to an abrupt interface, this gradient broadens the overlap of phonon density of states, thereby reducing phonon scattering and reflection, which can enhance interfacial thermal conductance despite the inherently low thermal conductivity of SiO_2 [56,57]. These results indicate that high-temperature annealing induces significant chemical reconstruction at the Si/PCD interface, leading to the formation of a compositionally graded interfacial region rather than an abrupt boundary. Such interfacial mixing is expected to enhance bonding stability by reducing

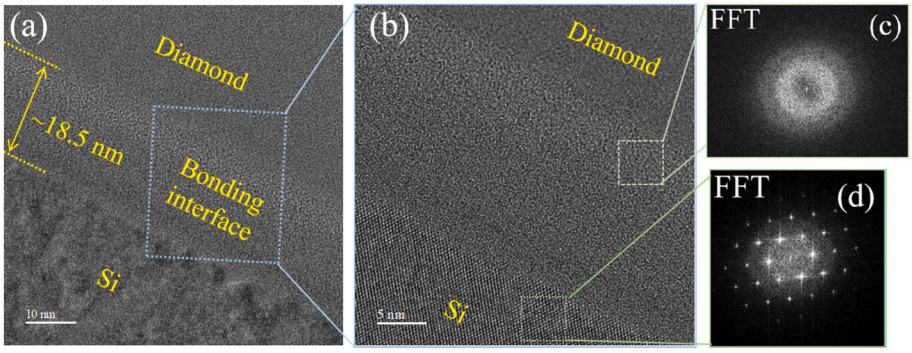


Fig. 5. (a) HR-TEM image and (b) corresponding magnified view of the as-bonded Si/PCD interface; (c, d) FFT patterns corresponding to the regions in the TEM images.

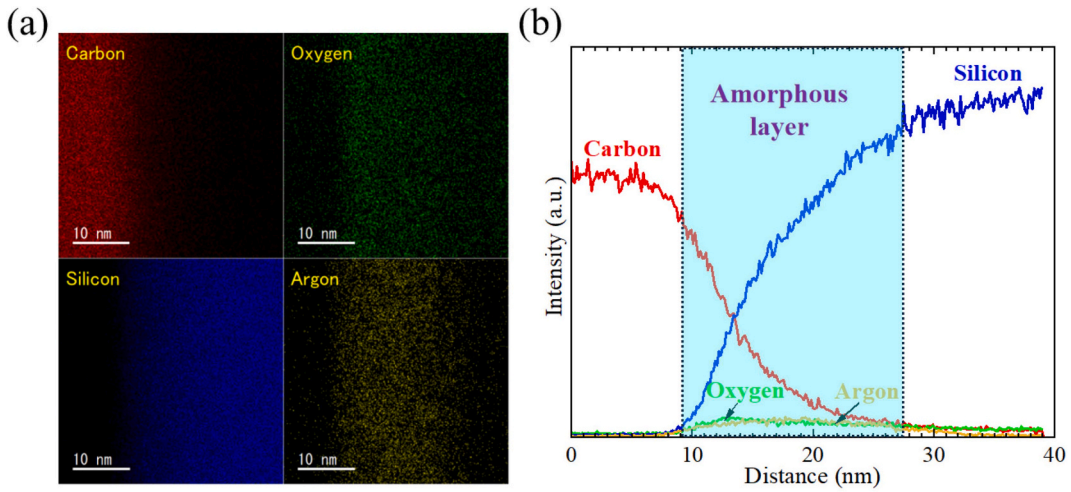


Fig. 6. (a) EDS elemental mapping of the as-bonded Si/PCD interface; (b) corresponding X-ray intensity profiles of carbon (red), oxygen (green), silicon (blue), and argon (yellow) across the interface.

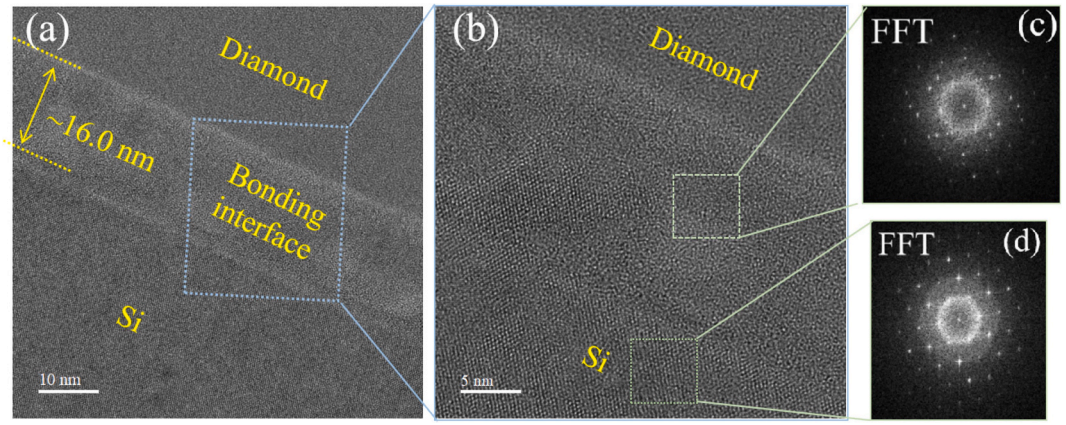


Fig. 7. (a) HR-TEM image and (b) corresponding magnified view of the Si/PCD interface after annealing at 1000 °C; (c, d) FFT patterns corresponding to the TEM images.

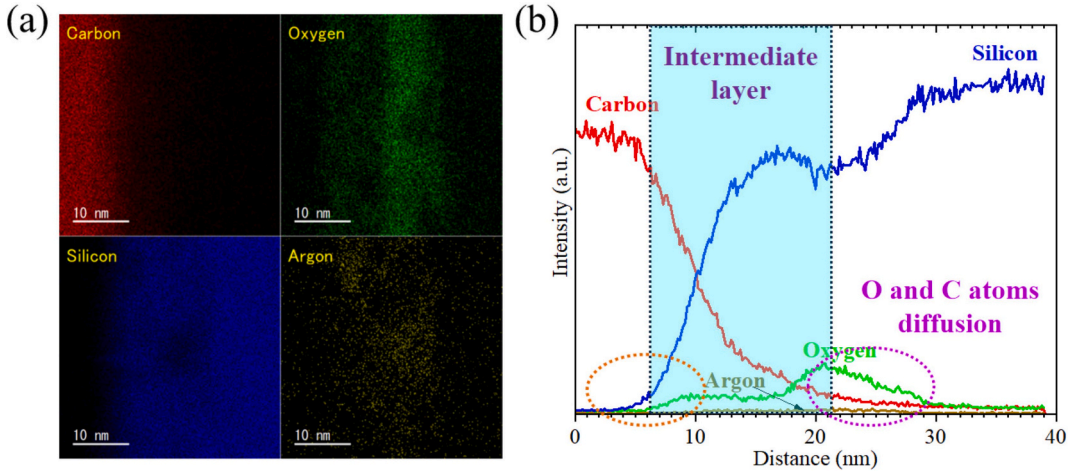


Fig. 8. (a) EDS elemental mapping of the Si/PCD interface after annealing at 1000 °C; (b) corresponding X-ray intensity profiles of carbon (red), oxygen (green), silicon (blue), and argon (yellow) across the interface.

chemical potential gradients and mitigating stress concentration. Meanwhile, the redistribution of light elements and the formation of a graded structure may substantially influence phonon transport across the interface, thereby affecting the TBC.

TDTR was employed to characterize the interfacial thermal transport behavior of the bonded samples. Pre-measurement thinning and

chemical polishing were implemented to satisfy the thickness and surface roughness requirements for TDTR measurements. No interfacial debonding or delamination was observed throughout sample preparation process, confirming that the bonded interface possesses sufficient mechanical strength to endure subsequent fabrication procedures. Fig. 9 (a) and 9(b) present the TDTR measurement results for ratio signals and the extracted TBC of the Si/PCD bonded samples as a function of annealing temperature. All annealing treatments were carried out under nitrogen atmosphere for 1 min, and three measurement points were tested for each specimen. A clear monotonic increase in TBC is observed with increasing annealing temperature, reaching a value of 133 MW m^{-2} K^{-1} after annealing at 1000 °C. Notably, this value approaches the theoretical value for the Si/PCD heterointerface reported in the reference [58], indicating the formation of a highly efficient thermal transport pathway across the interface. As summarized in Table 2, the TBC values obtained in this work surpass those reported in previous studies, demonstrating that the SAB-based heterogeneous integration achieves state-of-the-art interfacial thermal performance [28,30,35,46,59–61]. As previously discussed, Ar FAB irradiation during bonding disrupts the original lattice structure on both sides of the Si/PCD interface, generating a high density of reactive atomic sites to promote covalent bonding across the interface. This process concurrently forms a defect-rich amorphous interlayer with a thickness of approximately ~ 18.5 nm at the Si/PCD interface. This disordered layer plays a dual role in interfacial thermal transport. On one hand, it acts as a compliant buffer layer which enhances thermal stability by accommodating the large mismatch in CTE. On the other hand, the abrupt structural transition and the disordered amorphous layer strongly scatter phonons, impeding heat transport and resulting in a low TBC. A thinner amorphous layer generally exhibits insufficient thermal robustness, particularly under large CTE mismatch, whereas a thicker layer is helpful to improve stress accommodation at the expense of reduced TBC. With increasing annealing temperature, thermally activated atomic migration drives recrystallization of the interlayer, progressively reducing interfacial defects and the thickness of the amorphous layer. Due to recrystallization at 1000 °C, Si nanocrystals are formed within the interfacial region, rendering the two-phase junction continuous and smooth. This structural ordering substantially suppresses phonon scattering at the interface, leading to increase in TBC with annealing. This structural transition from an amorphous state to a partially ordered state not only enhances interfacial stability but also reduces phonon scattering, thereby facilitating more efficient heat transport and resulting in a higher TBC [62,63]. In addition, the presence of the intermediate layer plays a critical role in relieving residual stress induced by the CTE mismatch between Si and diamond [45]. These results reveal a fundamental trade-off between interfacial structural stability and thermal transport. Optimizing the thickness and structural evolution of the interfacial layer—particularly the amorphous-to-crystalline transition—is therefore essential for simultaneously achieving high thermal reliability and ultrahigh TBC in Si/diamond bonded systems.

4. Conclusions

1-inch PCD and 2-inch Si wafers were successfully bonded at room temperature using the SAB method. The bonded structures exhibit excellent thermal stability, maintaining structural integrity after annealing at temperatures up to 1000 °C. In the as-bonded state, an amorphous interfacial layer with a thickness of approximately 18.5 nm is observed, which transforms into a ~ 16 nm-thick Si nanocrystalline layer upon annealing at 1000 °C. Notably, no voids or interfacial defects are detected even after high-temperature treatment, indicating the formation of a robust and reliable heterointerface. Thermal characterization reveals that the TBC of the Si/PCD interface increases with annealing temperature, reaching a maximum value of 133 MW/ m^2 K at 1000 °C. In addition, the scalability and robustness of this bonding approach are demonstrated by successfully integrating 2-inch and 4-inch PCD wafers with 4-inch Si wafers. The bonded 4-inch PCD/Si wafers exhibit an ultralow bow value of 30 ± 5 μm, satisfying the requirements for wafer-scale semiconductor processing. These results establish SAB as a viable and scalable strategy for achieving high-quality Si/diamond heterogeneous integration. More importantly, this work elucidates the critical role of interfacial structural evolution—from an amorphous to a partially crystalline state—in simultaneously enhancing thermal stability and interfacial heat transport. By enabling high-TBC, thermally robust, and wafer-scale Si/diamond heterostructures, this approach provides a promising pathway for overcoming thermal management limitations in next-generation high-power and high-density electronic systems, and offers a general framework for engineering high-performance thermal interfaces in heterogeneous material systems.

CRediT authorship contribution statement

Liang Li: Writing – review & editing, Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Xiaoting Shen:** Visualization, Methodology, Data curation. **Zhe Cheng:** Validation, Methodology, Data curation. **Yutaka Ohno:** Resources, Formal analysis, Data curation. **Ke Xu:** Supervision, Resources, Formal analysis. **Xiong Zhang:** Writing – review & editing, Methodology, Investigation, Funding acquisition, Data curation. **Shengwang Yu:** Resources, Methodology, Data curation. **Ke Zheng:** Data curation, Methodology, Validation. **Jianbo Liang:** Writing – review & editing, Visualization, Resources, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

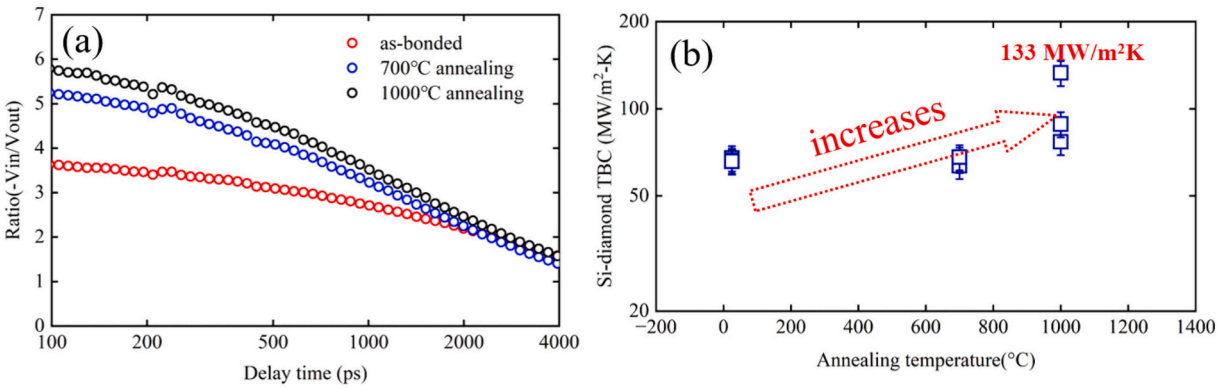


Fig. 9. (a) TDTR ratio signals and (b) extracted TBC of Si/PCD bonded samples as a function of annealing temperature.

Table 2
Summary of Si-Diamond TBC values reported in the literature and obtained in this work.

Data source	Method	Material	Specifications	TBC (MW/m^2K)	Conditions
Ref [28]	Nanocrystal diamond			50	
Ref [30]	Nanoscale Graphoepitaxy			105	
Ref [35]	Atomic diffusion bonding	Single-crystal diamond/Si	5×5 mm^2	41	200℃
Ref [46]	Atomic diffusion bonding	Polycrystalline diamond/Si		102	200℃
Ref [59]	Surface activated bonding	Single-crystal diamond/Si	10.3×10.3 mm^2	105	room temperature
Ref [60]	Ag sinter pastes	Polycrystalline diamond/Si	27×28 mm^2	23	> 200℃
Ref [61]	CVD diamond on Si			67	
This work	Surface activated bonding	Polycrystalline diamond/Si	1-, 2, 4-inch	133	room temperature

the work reported in this paper.

Acknowledgments

This work was partially supported by JSPS KAKENHI (Grant No. JP24K01364), NSFC (Grant No. T2550270), the Key Research and Development Project of Science and Technology Department of Jiangsu Province, The People's Republic of China (Grant No. BE2021008-4), and the Postgraduate Research & Practice Innovation Program of Jiangsu Province (Grant No. KYCX25_0435).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apsusc.2026.167977.

Data availability

Data will be made available on request.

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